Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHNOLOGY

(Autonomous Institute Affiliated to University of Mumbai)

Munshi Nagar, Andheri (W), Mumbai – 400 058.

End Semester Examination

Dec. 2024

Max. Marks: 100 | **Duration: 180 Min.**

Class: Final Year BTech(CE) | **Semester: VII**

Course Code: CS401 | **Program: BTech(CE)**

Name of the Course: Cryptography and System Security

Instruction:

1) **All questions are compulsory.**
2) **Draw neat diagrams and write equations wherever necessary.**
3) **Assume suitable data if necessary but justify the same.**

Course Outcomes: At the end of the course, student will be able to:

(1) Describe the different types of the cryptographic algorithms to secure information.
(2) Apply different cryptographic techniques to solve security-related problems
(3) Create a message digest from data to authenticate authorized user
(4) Use system security practices

Q. No.	Question	Max. Marks	CO-BL
Q. 1A	How many possible keys does the Playfair cipher have? Ignore the fact that some keys might produce identical encryption results. Express your answer as an approximate power of 2 with a detailed justification. Now take into account the fact that some Playfair keys produce the same encryption results. How many effectively unique keys does the Playfair cipher have? Justify the same.	**10**	CO1-BL5
Q. 1B	a) Encrypt the message "meet me at the usual place at ten rather than eight oclock" using the Hill cipher with the key $\begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$. Show your calculations and the result. b) Show the calculations for the corresponding decryption of the ciphertext to recover the original plaintext.	**10**	CO1-BL3
Q. 2A	This problem illustrates the point that the Diffie-Hellman protocol is not secure without the step where you take the modulus; i.e. the "Indiscrete Log Problem" is not a hard problem! You are Eve, and have captured Alice and Bob and imprisoned them. You overhear the following dialog. **Bob:** Oh, let's not bother with the prime in the Diffie-Hellman protocol, it will make things easier. **Alice:** Okay, but we still need a base to raise things to. How about g = 3? **Bob:** All right, then my result is 27. **Alice:** And mine is 243. What is Bob's secret X_B and Alice's secret X_A? What is their secret combined key? (Don't forget to show your work/calculations in detail.)	**10**	CO1-BL3
Q. 2B	In an RSA system, the public key of a given user is e = 31, n = 3599. What is the private key of this user? Hint: First use trial and error to determine p and q; then use the extended Euclidean algorithm to find the multiplicative inverse of 31 modulo $\phi(n)$.	**10**	CO2-BL3
Q. 3A	Discuss Kerberos protocol and show all message exchanges of three servers in Kerberos Protocol. Explain how the Kerberos protocol ensures secure authentication in a distributed system. What are the roles of the Ticket Granting Ticket (TGT) and session keys in the process?	**10**	CO2,3-BL4

Q. 3B	What characteristics are needed in a secure hash function? Show each characteristic formally in terms hash function H, hash value H(x) for input x etc. Discuss at least five characteristic in detail. **OR** Show the message digest generation using SHA-512 in detail. Compare different version of Secure Hash Algorithm (SHA) in terms of SHA parameters.	**10**	CO3,4-BL4
Q. 4A	a) Bob receives a PGP message. How can he find out the type of the packet if the tag value is 8, 9 and 2. Disucss each packet in detail. b) In PGP, can an e-mail message use two different public-key algorithms for encryption and signing? How is defined in a message sent from Alice to Bob?	**10**	CO2,3, 4-BL3
Q. 4B	a) What are three benefits that can be provided by an intrusion detection system? b) What is the difference between rule-based anomaly detection and rule-based penetration identification?	**10**	CO4-BL2
Q. 5A	Draw a diagram and show actual ISAKMP packets that are exchanged between an initiator and a responder using the preshared-key method in the main mode. Use at least two proposal packets with at least two transform packets for each proposal.	**10**	CO3,4-BL3
Q. 5B	What are the length of the key-material (client and server authentication keys, client and server encryption keys, client and server IV) and the number of repeated modules needed, if the cipher suites are as follows: a) SSL_RSA_WITH_NULL_MDS b) SSL_RSA_WITH_NULL_SHA c) TLS_RSA_WITH_DES_CBC_SHA d) TLS_RSA_WITH_3DES_EDE_CBC_SHA e) TLS_DHE_RSA_WITH_DES_CBC_SHA f) TLS_DH_RSA_WITH_3DES_EDE_CBC_SHA	**10**	CO3,4-BL3